



Data Communication Wireless Networks-IE2100

2nd Year, 2nd Semester

Group Assignment

Submitted to

Sri Lanka Institute of Information Technology

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Bachelor of Science Special Honors Degree in Information Technology

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Introduction

Behpaya Information Technology is a professional services company with approximately 50 staff operating from a two-floor head office in Sri Lanka. The office includes open-plan work areas, meeting rooms, a boardroom, reception and a café/lobby area. Current connectivity relies on a single 300 Mbps ISP fiber link and mixed legacy devices with limited centralized management.

In response to the invitation, our group submits this proposal for the design, supply, installation, configuration, testing, documentation and support of a modern wired and wireless network. The objective is to deliver a secure, reliable and easily managed network that supports current operations and anticipates growth over the next 3-5 years.

Key Requirements

1. Wired LAN: Managed PoE+ switches, VLAN segmentation, QoS for voice/video.
2. Wireless LAN: Wi-Fi 6 Aps, controller-based management, roaming optimization, 65 dBm coverage.
3. Security: Next-generation firewall with IPS/AV/URL filtering, VPN with MFA, basic NAC.
4. Cabling: Cat6/Cat6a structured cabling with certification.
5. Management: IP/VLAN planning, DHCP/DNS, monitoring, backups and RBAC.

Physical & Logical Layout of Behpaya Information Technology

1. Physical Layout of Behpaya Information Technology

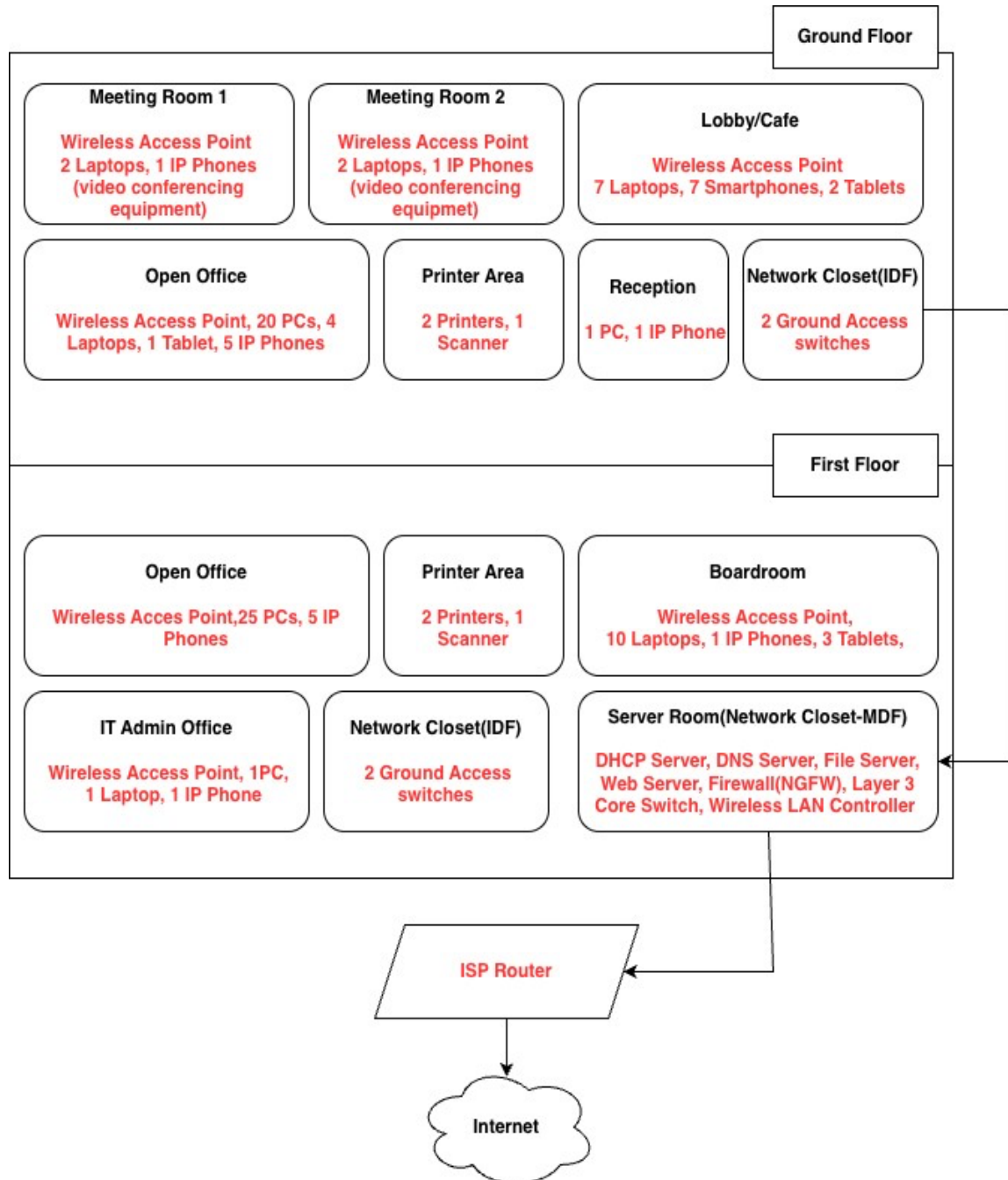


Figure 1.0

2. Logical Layout of the Behpaya Information Technology

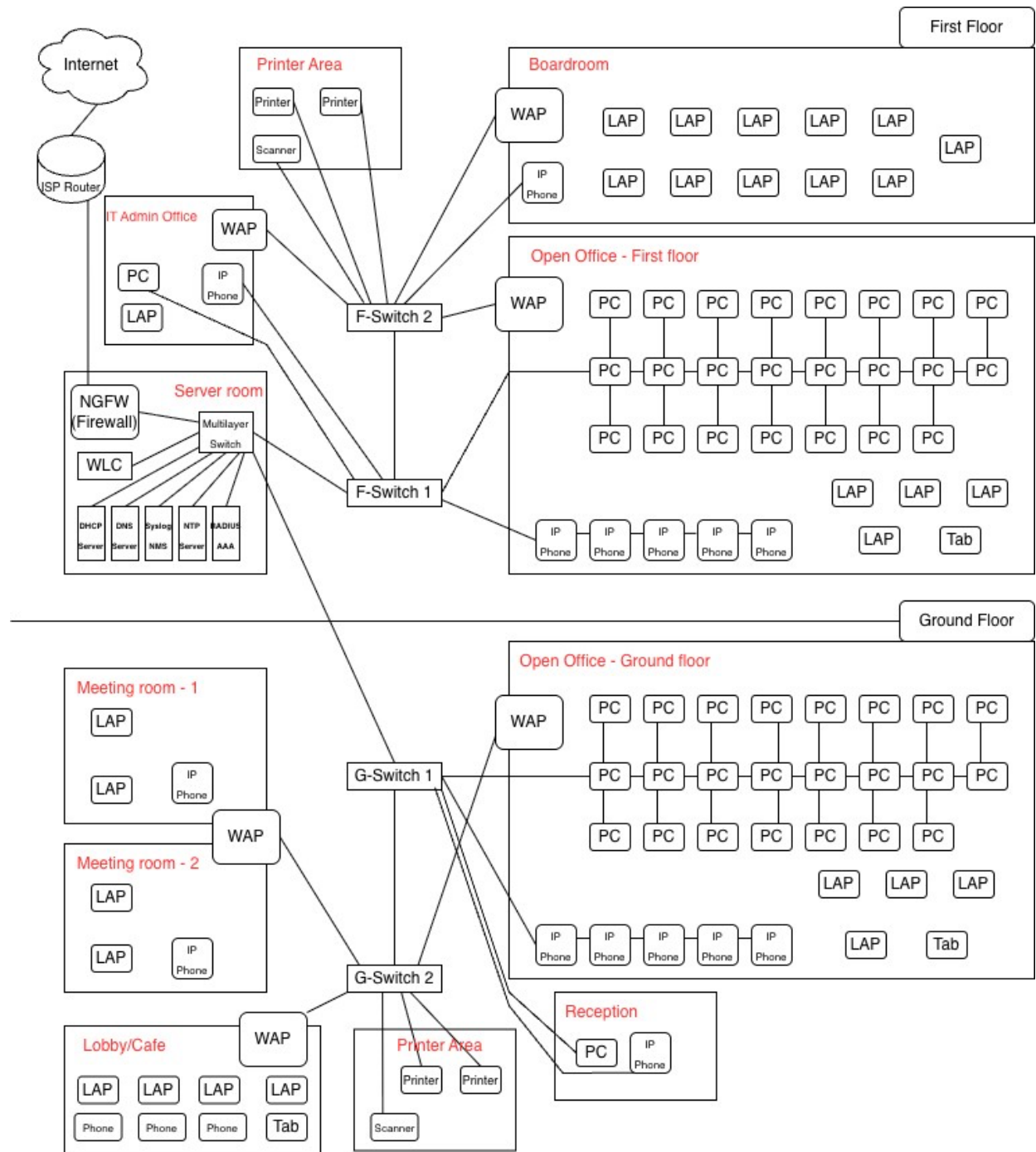


Figure 2.0

3. Logical layout overview for Behpaya Information Technology using Cisco Packet Tracer

Ground Floor:

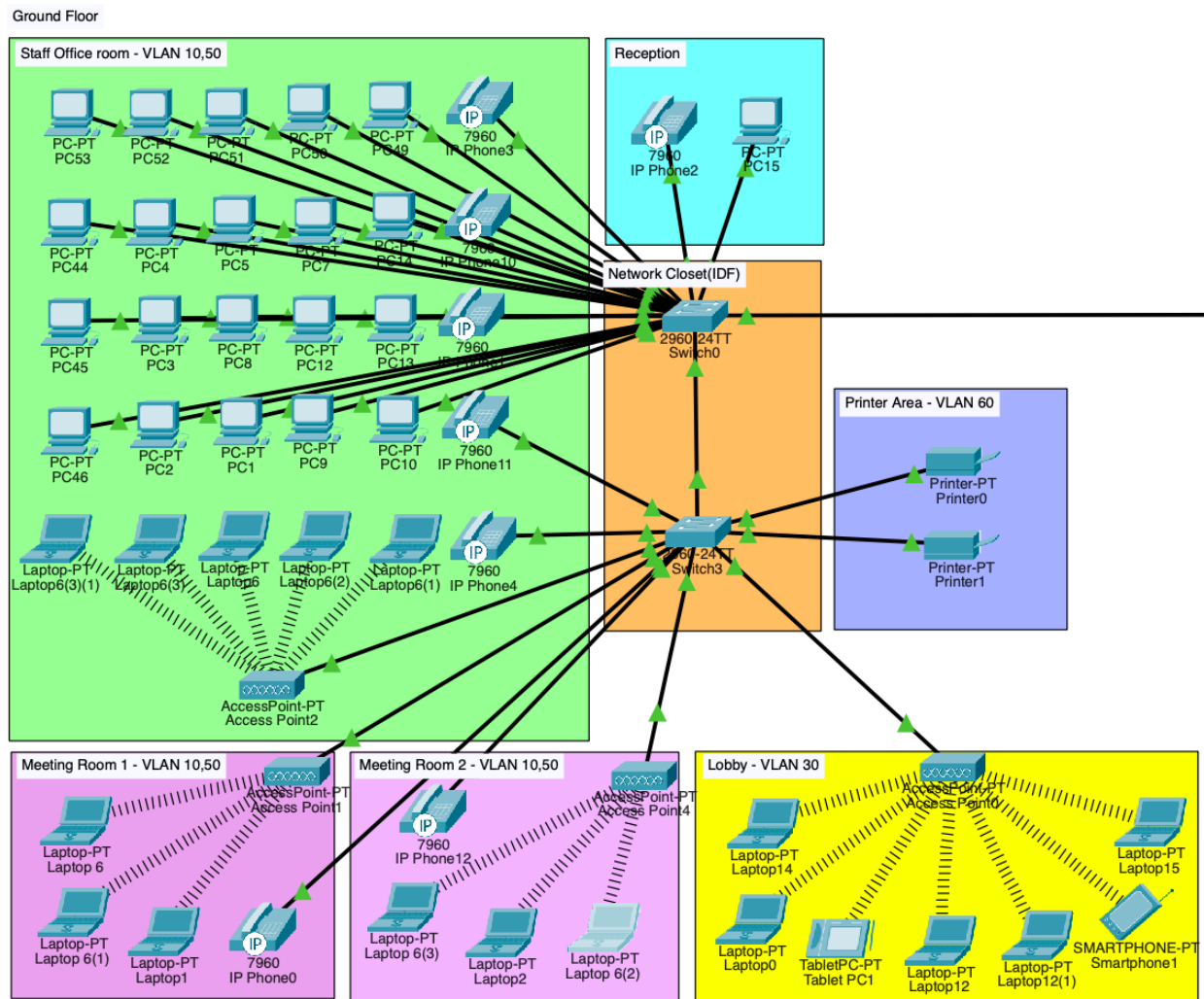


Figure 3.1

First Floor:

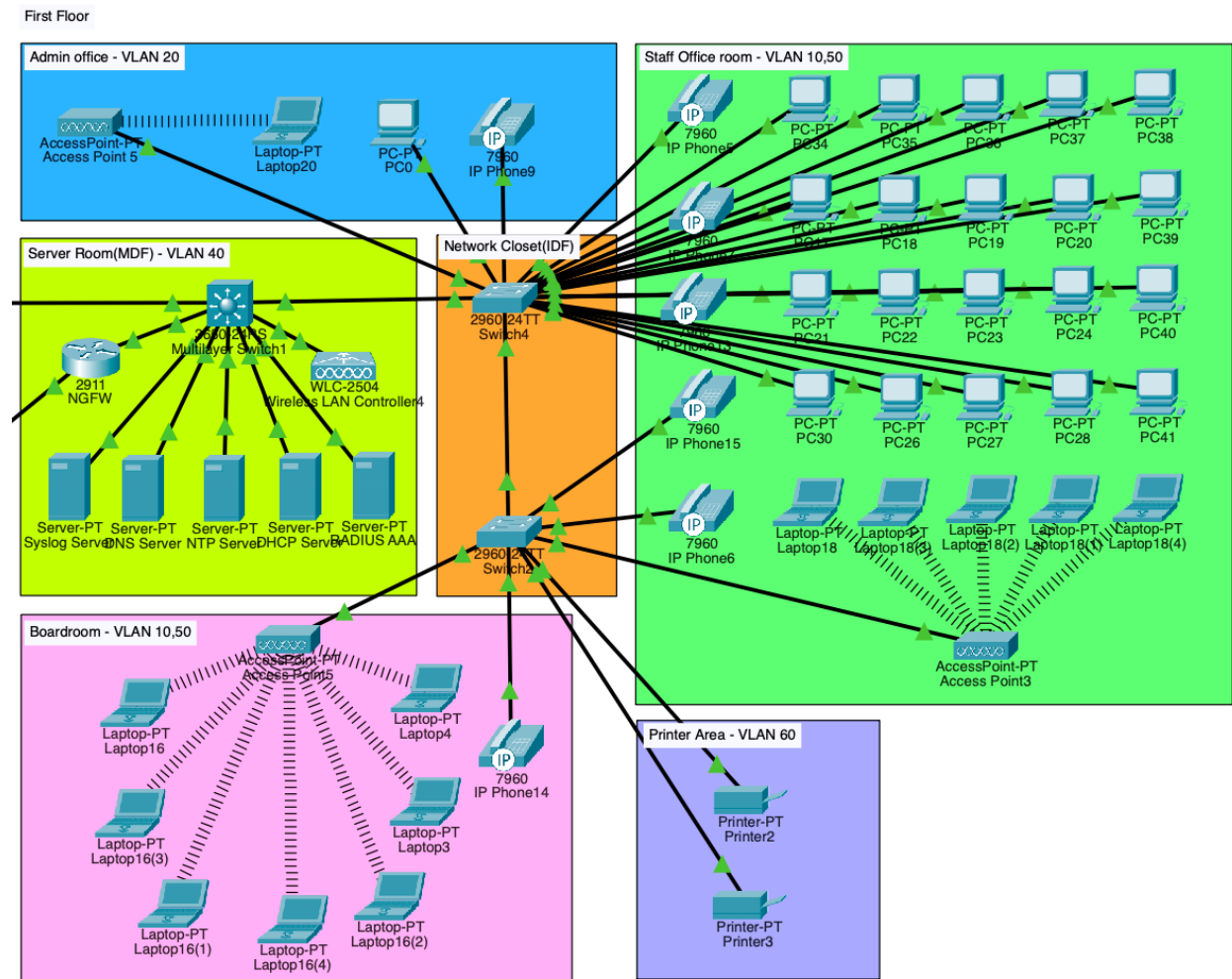


Figure 3.2

4. Floor Plan of the Bahpaya Information Technology

Ground Floor:

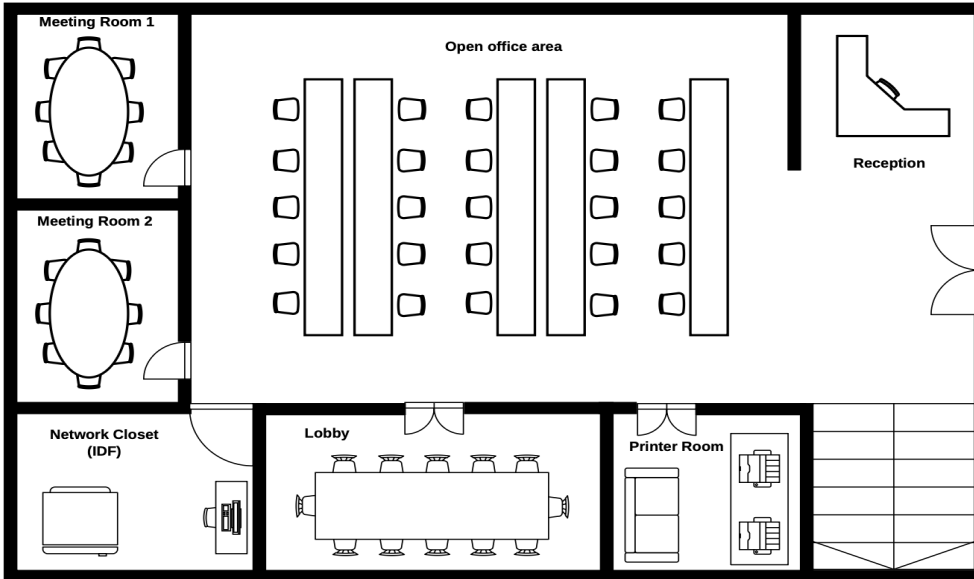


Figure 4.1

First Floor:

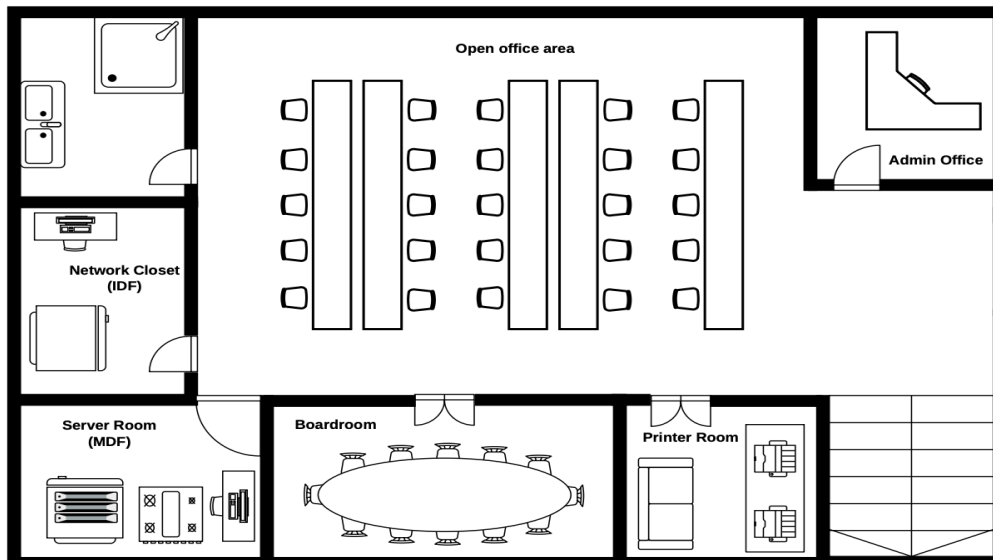


Figure 4.2

5. Topology Description

Layer	Device	Quantity
Edge Layer	NGFW Firewall	1
Core Layer	Layer 3 Switch (L3-CORE)	1
Distribution/Access Layer	Layer 2 Access Switches	4

6. Device Connections

From Device	Port	To Device	Port
L3-CORE	Gig0/1	NGFW	Gig0/1
L3-CORE	Gig0/2	SW-G1	Gig0/1
L3-CORE	Gig0/3	SW-F1	Gig0/1
SW-G1	Gig0/2	SW-G2	Gig0/2
SW-F1	Gig0/2	SW-F2	Gig0/2

Project Management & Proposal

1. Project Methodology

This project will follow the 'PDIOO methodology' (Plan, Design, Implement, Operate, Optimize)

Phase	Activities	Duration
Plan	Requirements gathering, site survey, stakeholder interviews	3 days
Design	Low-Level Design (LLD), VLAN planning, heatmaps, Bill of Materials	5 days
Implement	Structured cabling, switch configuration, AP mounting, firewall deployment	10 days
Operate	Functional testing, acceptance testing, user training, handover	2 days
Optimize	Fine-tuning RF power, documentation finalization, knowledge transfer	1 day

2. Resource Allocation

Role	Team Member	Responsibilities	Contact
Project Manager	Bandara K.M.G.A.P.	Timeline management, client communication, final sign-off	it24102705@my.sliit.lk
Network Engineer (Wired)	Fernando C.J.S.	Switch configuration, VLAN implementation, cabling supervision	it24103002@my.sliit.lk
Network Engineer (wireless)	Jayasena W.M.S.A.	AP placement, SSID design, roaming optimization, heatmap validation	it24100816@my.sliit.lk
Security Specialist	Dissanayake N.K.	Firewall configuration, VPN setup, RADIUS/802.1X deployment	it24100031@my.sliit.lk

Implementation Plan

1. Project Timeline

Week	Activities
Week 1	Site survey, cabling installation, rack mounting
Week 2	Switch configuration, VLAN setup, trunking
Week 3	AP installation, wireless configuration, firewall deployment
Week 4	Testing, validation, user training, handover

2. Risk Management

Risk	Probability	Impact	Mitigation
Cabling delays	Medium	High	Order materials 2 weeks in advance
Configuration errors	Low	Medium	Backup configs before changes
AP coverage gaps	Low	Medium	Conduct predictive heatmap first
Staff resistance	Low	Low	Provide training and documentation

Wired LAN Design

1. Switch Port Allocation

Ground Floor – Switch 1(SW-G1)

Port Range	VLAN	Device Type	Quantity
Fa0/1-20	10 (STAFF)	Staff PCs	20
Fa0/21-22	20 (ADMIN)	Admin PCs	2
Fa0/23-24	50 (VOIP)	IP Phones	2
Gig0/1	Trunk	To L3 Switch	-
Gig0/2	Trunk	To Ground SW2	-

Ground Floor – Switch 2 (SW-G2)

Port Range	VLAN	Device Type	Quantity
Fa0/1-5	10 (STAFF)	Staff PCs	5
Fa0/6-15	50 (VOIP)	IP Phones	10
Fa0/16-19	70 (AP_MGMT)	APs	3
Fa0/20-21	60 (PRINTERS)	Printers	2
Fa0/22-24	30 (GUEST)	Guest Wired	3
Gig0/2	Trunk	To Ground SW1	-

First Floor – Switch 1 (SW-F1)

Port Range	VLAN	Device Type	Quantity
Fa0/1-20	10 (STAFF)	Staff PCs	20
Fa0/21-22	20 (ADMIN)	Admin PCs	2
Fa0/23-24	50 (VOIP)	IP Phones	2
Gig0/1	Trunk	To L3 Switch	-
Gig0/2	Trunk	To First SW2	-

First Floor – Switch 2 (SW-F2)

Port Range	VLAN	Device Type	Quantity
Fa0/1-5	10 (STAFF)	Staff PCs	5
Fa0/6-15	50 (VOIP)	IP Phones	10
Fa0/16-19	70 (AP_MGMT)	APs	3
Fa0/20-21	60 (PRINTERS)	Printers	2
Fa0/22-24	30 (GUEST)	Guest Wired	3
Gig0/2	Trunk	To First SW1	-

2. Key Configuration Features

Feature	Implementation
VLAN Segmentation	7 VLANs (Staff, Admin, Guest, Mgmt, VoIP, Printers, IoT)
Trunking	802.1Q trunk between switches
Port Security	Max 2 MAC addresses per port, restrict violation
DHCP Snooping	Enabled on all user VLANs (Except VLAN 40)
Spanning Tree	Portfast on access ports
Link Aggregation	2x1 Gbps uplinks (LACP)

Wireless LAN Design

GROUND FLOOR - 5 GHz HEATMAP

Coverage: 96% at -65 dBm ($\geq 95\%$ requirement met)



Figure 5.1

GROUND FLOOR - 2.4 GHz HEATMAP

Coverage: 98% at -65 dBm ($\geq 95\%$ requirement met)

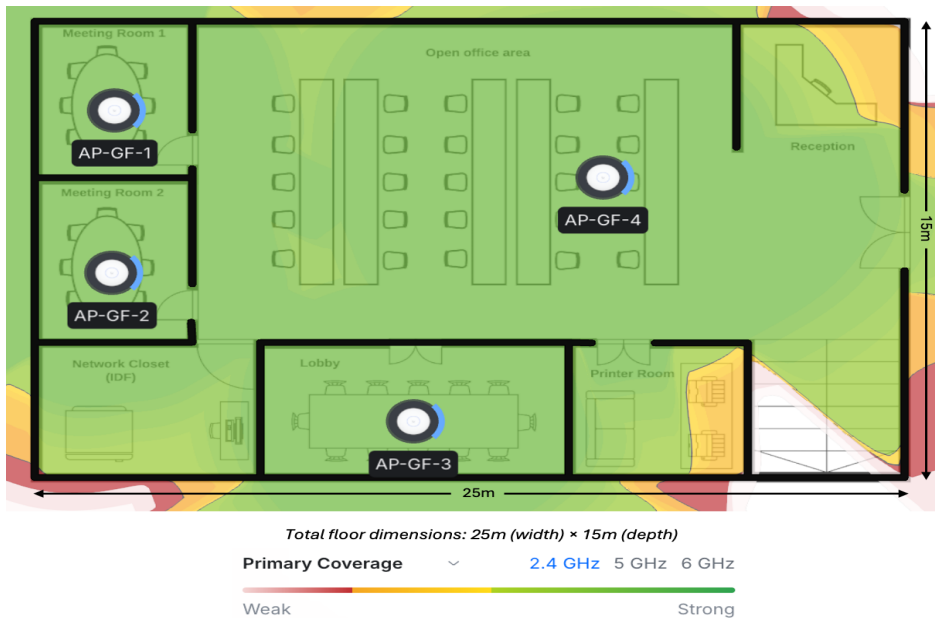


Figure 5.2

First FLOOR - 5 GHz HEATMAP

Coverage: 96% at -65 dBm ($\geq 95\%$ requirement met)



Figure 5.3

GROUND FLOOR - 2.4 GHz HEATMAP

Coverage: 98% at -65 dBm ($\geq 95\%$ requirement met)

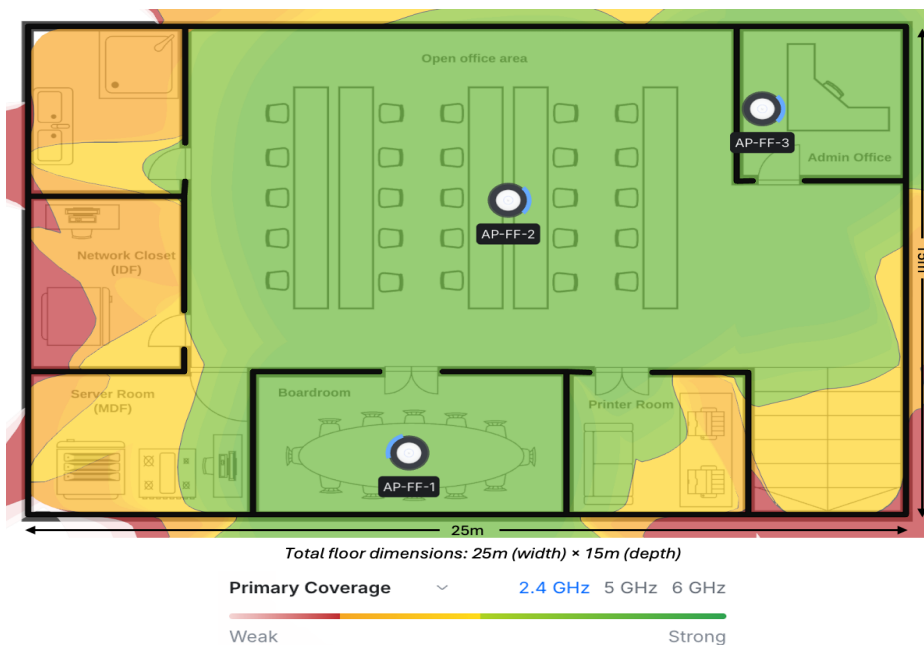


Figure 5.4

1. SSID Design

SSID	VLAN	Authentication	Encryption	Band	Purpose
Behpaya_Corp	10	802.1X (RADIUS)	WPA3-Enterprise	5/6 GHz	Staff laptops, phones
Behpaya_Admin	20	802.1X + MFA	WPA3-Enterprise	5 GHz	IT/admin devices
Behpaya_Guest	30	Captive Portal	WPA2-Personal	2.4/5 GHz	Guest internet access

2. AP Placement Plan

Floor	AP ID	Location	Coverage Area
Ground	AP-G1	Reception	Reception area
Ground	AP-G2	Open Office	5 staff desks
Ground	AP-G3	Meeting Room 1	2 meeting rooms
First	AP-F1	Open Office	5 staff desks
First	AP-F2	IT Admin Office	Admin desk
First	AP-F3	Boardroom	Boardroom

3. Coverage Targets

Metric	Target
Minimum Signal	-65 dBm (5 GHz)
Coverage Area	≥95% of office space
Roaming Handoff	<150 ms ⁻¹
Channel Width	20/40 MHz (auto)
Transmit Power	Auto adjusted by controller

RF Planning & Channel design

1. Channel Plan (5 GHz)

Floor	AP ID	Channel	Width	Power (dBm)	Coverage area
Ground	AP-G1	36	40 MHz	17	Reception
Ground	AP-G2	40	40 MHz	20	Open Office
Ground	AP-G3	44	20 MHz	14	Meeting Office
first	AP-F1	48	40 MHz	20	Open Office
First	AP-F2	52	20 MHz	14	Admin Office
Firs	AP-F3	56	20 MHz	17	Boardroom

2. Channel Plan (2.4 GHz)

Floor	AP ID	Channel	Width	Power (dBm)
Ground	AP-G1	1	20 MHz	11
Ground	AP-G2	6	20 MHz	14
Ground	AP-G3	11	20 MHz	8
First	AP-F1	1	20 MHz	14
First	AP-F2	6	20 MHz	8
First	AP-F3	11	20 MHz	11

Security Edge Design

1. Firewall Features

Feature	Implementation
Stateful Throughput	1 Gbps
Threat Protection	300 Mbps
IPS/IDS	Enabled with daily signature updates
URL Filtering	80+ categories
Anti-Virus	Gateway AV scanning
Geo-IP Blocking	Block high-risk countries

2. VPN Configuration

VPN Type	Users	Authentication
SSL VPN	Remote staff (up to 20 concurrent)	MFA (SMS/App)
Site-to-Site IPsec	Future branches	Pre-shared keys

3. Threat Identification & Mitigation

Threat	Likelihood	Impact	Mitigation	Residual Risk
Unauthorized access to wired network	Medium	High	802.1X + MAC filtering	Low
Evil twin AP attack	Medium	High	Rogue AP detection (WIDS)	Low
DoS attack on guest Wi-Fi	High	Medium	Rate limiting (10 Mbps/client)	Medium
Man-in-the-middle attack	Low	High	ARP inspection + DAI	Low
Data breach via guest network	Medium	Medium	L3 isolation + URL filtering	Low
Weak Wi-Fi password cracking	Medium	High	WPA3-Enterprise + MFA	Low
Physical switch port tampering	Low	Medium	Port security + disable unused ports	Low

4. Access Control Policies

Source	Destination	Action
Staff (VLAN 10)	Internet, Internal	Allow
Admin (VLAN 20)	All	Allow
Guest (VLAN 30)	Internet Only	Allow (block internal)
VoIP (VLAN 50)	Internet (SIP)	Allow with QoS
Printers (VLAN 60)	Internal only	Allow

5. 802.1X Authentication

Authentication flow:

- i. Device connects to switch port or SSID.
- ii. Switch/AP requests credentials.
- iii. RADIUS server (192.168.40.14) authenticates.
- iv. VLAN assigned based on user role.
- v. Device granted network access.

Network Management System (NMS) Justification

1. Chosen Tools

Tool	Purpose	Reason to choose
Cisco Packet Tracer	Simulation	Free, educational, supports full config
SNMP (SolarWinds/PRTG)	Monitoring	Industry standard, threshold alerts
Syslog Server (Kiwi)	Log aggregation	Centralized troubleshooting
NetFlow (nTop)	Bandwidth analysis	Identify top talkers

2. NMS Feature Mapping Requirements

Requirement	NMS Feature	How It's met
Centralized monitoring	SNMP traps	All devices send traps to 192.168.40.12
Configuration backup	SCP/TFTP	Daily backups to NMS server
Threshold alerts	SNMP polling	Email when CPU >80%, WAN down
Role-based access	RBAC	Admin (full), Operator (view only)
Change control	Config versioning	Before/after configs archived

Network Management

1. Server Configuration

Server	IP Address	Function
DHCP Server	192.168.40.10	Dynamic IP assignment
DNS Server	192.168.40.11	Name resolution
Syslog/NMS	192.168.40.12	Logging & monitoring
NTP Server	192.168.40.13	Time synchronization
RADIUS/AAA	192.168.40.14	802.1X authentication

2. Monitoring & Alerts

Alert Type	Threshold	Action
WAN Down	Immediate	Email to admin
AP Offline	>5 minutes	Email + Ticket
Switch Port Error	>10 errors/min	Email alert
High CPU (Firewall)	>80%	Email alert

3. Backup Procedure

Item	Frequency	Retention
Switch configs	Daily	30 days
Firewall config	Daily	90 days
Controller backup	Weekly	6 months

IP Addressing & VLAN Plan

1. VLAN Summary

VLAN ID	Name	Subnet	DHCP	Gateway	Usable IPs
10	STAFF	192.168.10.0/24	Yes	192.168.10.1	254
20	ADMIN	192.168.20.0/28	Yes	192.168.20.1	14
30	GUEST	172.16.30.0/24	Yes	172.16.30.1	254
40	NETWORK_DEV	192.168.40.0/28	No	192.168.40.1	14
50	VOIP	192.168.50.0/24	Yes	192.168.50.1	254
60	PRINTERS	192.168.60.0/28	Yes	192.168.60.1	14
70	AP_MGMT	192.168.70.0/24	Yes	192.168.70.1	254
99	IOT	192.168.99.0/24	Yes	192.168.99.1	254

2. DHCP Scopes

VLAN	Pool Range	Lease Time
10	192.168.10.100 - 192.168.10.200	8 hours
20	192.168.20.10 - 192.168.20.25	8 hours
30	172.16.30.50 - 172.16.30.200	4 hours
50	192.168.50.50 - 192.168.50.150	24 hours
60	192.168.60.10 - 192.168.60.20	24 hours
99	192.168.99.10 - 192.168.99.100	12 hours

Bill Of Materials (BOM)

Item	Model	Qty	Unit Price (LKR)	Total (LKR)
NGFW Hardware	FortiGate 80F	1	450,000	450,000
NGFW Software (3-Yr)	FortiGuard Security Bundle	1	270,000	270,000
L3 Core Switch	Cisco 3560-24PS	1	350,000	350,000
Access Switch	Cisco 296024TT	4	120,000	480,000
Access Switch Software	Cisco ONE Foundation (24 ports)	4	25,000	100,000
Wi-Fi AP	Meraki MR46	6	85,000	680,000
Cloud Controller License (3-Yr)	Meraki Cloud MR Advanced Security	6	52,000	416,000
Cat6A Cabling	305m box	3	25,000	75,000
Patch Panel	24-port	4	8,000	32,000
Rack	42U Server Rack	1	60,000	60,000
UPS	APC SMT1500	1	85,000	85,000
Installation & Configuration	-	1	300,000	300,000
Total Estimated Cost	3,298,000 LKR			

Testing & Acceptance

1. Test matrix

Test	Method	Pass Criteria
Wired Connectivity	Ping test	100% success
DHCP Assignment	ipconfig on PC	Correct IP per VLAN
VLAN Isolation	Ping across VLANs	Guest → Staff = Fail
Wireless Coverage	Site survey tool	≥95% @ -65 dBm
Roaming	Continuous ping while walking	<2 packet loss
Internet Access	Browse websites	Success
Firewall Rules	Attempt blocked sites	Blocked
VPN Connection	Remote users connect	Success with MFA

2. Acceptance & Sign-Off

Client Representative

Full Name:

Signature:

Date:

Vendor Representative

Full Name:

Signature:

Date:

Training & Documentation

1. Deliverables

Item	Format	Recipient
Low-Level Design (LLD)	PDF/Word	IT Manager
As-built Network Diagrams	Visio/PDF	IT Team
Device Inventory	Excel	IT Team
Configuration Backups	Encrypted file	IT Manager
Admin Quick-Start Guide	PDF	IT Team

2. Training Session

Duration: 2-4 hours

Topics:

- Daily Networking monitoring
- Adding/removing users
- Guest Wi-Fi management
- Backup and restore procedures
- Basic troubleshooting

Conclusion

This proposal has represented a comprehensive, enterprise-grade network design for Behpasa Information Technology's two floor head office. The solution directly addresses every requirement outlined in the RFP while delivering additional value through thoughtful design and industry best practices.

How requirements have been met:

RFP Requirement	Proposed Solution
Managed PoE+ switches	Cisco 2960-24TT with VLAN segmentation & port security
Wi-Fi 6 coverage	8 x Meraki MR46 APs with cloud controller
95% coverage -65 dBm	Predictive heatmaps with 4 APs per floor
Guest isolation	Dedicated VLAN 30 with ACL blocking internal access
NGFW security	FortiGate 80F with IPS/AV/URL filtering & VPN
Centralized management	Syslog server, SNMP alerts, daily config backups

Business Value:

This Solution delivers significant benefits beyond technical compliance. Employees will experience reliable connectivity with minimal downtime, directly improving productivity. The NGFW and 802.1X authentication framework significantly reduce security risks compared to the current legacy environment. Guest Wi-Fi with captive portal enhances the company's professional image when clients visit.

Scalability & Future Growth:

The design accommodates the current 50 staff and 30 concurrent guests while supporting growth to 150 staff over the next 3-5 years without major hardware upgrades. Stackable switches, cloud-managed Wi-Fi and modular VLAN design ensure easy expansion.

Conclusion & Recommendation:

All RFP requirements have been met or exceeded. The solution is secure, reliable, cost-effective and can be fully implemented within 4 weeks. We strongly recommend that Behpasa Information Technology approve this proposal and proceed with implementation.

References

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